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DIFFERENTIAL GEOMETRY CHAPTER #3 21/04/2026

Q1 A point P on a surface is called hyperbolic if the gaussian curvature K at P is negative This means that the two principle curvatures have opposite signs An asymptotic direction ~~is a dir~~ at a point P is a direction vector v for which $k_n = 0$. (k_n is the normal curvature)

The normal curvature K(v) in any direction v can be expressed using Euler's formula:

$$k_n(v) = k_1\cos^2\theta + k_2\sin^2\theta$$

For an asymptotic direction, $k_1\cos^2\theta + k_2\sin^2\theta = 0$

$k_1\cos^2\theta = -k_2\sin^2\theta$ (dividing b/s by $\cos\theta$ since $\cos\theta \neq 0$ as $\theta = \frac{\pi}{2}$ means $k_2 = 0$ which is not true for a hyperbolic point

$$\tan^2\theta = -\frac{k_1}{k_2} \iff \tan^2\theta = m^2 \iff \theta_1 = \arctan(m) \text{ or } \theta_2 = \arctan(-m) = -\arctan(m)$$

let $m^2 = -\frac{k_1}{k_2}$

The principal directions are orthogonal to each other (e_1 and e_2). The angle between e_1 and first asymptotic direction is θ_1 & The angle between e_1 and second asymptotic direction is θ_2

Now consider the second principal direction e_2. The angle between e_2 and first asymptotic direction is $|\frac{\pi}{2} - \theta_1|$ & angle between e_2 and second asymptotic direction is $|\frac{\pi}{2} - (-\theta_1)| = |\frac{\pi}{2} + \theta_1|$

For the principle directions to bisect the asymptotic directions, each principle direction must lie on the angle bisector of the two asymptotic direction

- The asymptotic directions are at angle θ_1 & $-\theta_1$ relative to e_1
- The asymptotic directions are at angle $|\frac{\pi}{2} - \theta_1|$ & $|\frac{\pi}{2} + \theta_1|$ relative to e_2.

Q2: Let c be a curve in S such that, for every point in C, such that s is tangent to a plane T. Let us consider $\alpha : (-\varepsilon, \varepsilon) \to C \subset S$ a regular parametrisation of C such that $\alpha(0) = p \in S$, $\alpha'(0) = v \in T_pS$, $\|\alpha\| = 1$. Since S is tangent to the plane t at every $\alpha(t) \in C$,

$$N(t) = (N \circ \alpha(t)) = N_0 \quad \forall t \in (-\varepsilon, \varepsilon)$$

So $N'(t) = dN_{\alpha(t)}(\alpha'(t)) = 0 \quad \forall t \in (-\varepsilon, \varepsilon)$

$dN_p(v) = 0 = 0v \Rightarrow k_1 = 0$ or $k_2 = 0 \Rightarrow \det(dN_p) = 0$ so it is either parabolic or planar

$dN_p = 0$

3. We know the normal curvature is given by:

$$k_n = k\langle n, n\rangle = k\cos\theta \Rightarrow |k_n| < |k|$$

$$k = |k| \geq$$

$$\pm k_1\cos^2\theta \pm k_2\sin^2\theta \leq k$$

$$|k_1|\cos^2\theta + |k_2|\sin^2\theta \leq k$$

$$k \geq \{\min\{|k_1|, k_2|\}\}(\cos^2\theta + \sin^2\theta)$$

$$k \geq \min\{|k_1|, |k_2|\}$$

Q4. Let's consider a circle on a plane P. At P, we have $k_1 = k_2 = 0$ which satisfies $|k_1| \leq 1$ & $|k_2| \leq 1$. This circle is a closed curve on S such that curvature $k = \frac{1}{r}$ which is > 1 for $r < 1$ which does not satisfy $|k| \leq 1$

Q5. $H = \frac{k_1 + k_2}{2}$ & $k_n = k_1\cos^2\theta + k_2\sin^2\theta$

$$\int_0^{\pi} k_n\,d\theta = \int_0^{\pi}(k_1\cos^2\theta + k_2\sin^2\theta)\,d\theta = \frac{k_1}{2}\int_0^{\pi}(1+\cos 2\theta)\,d\theta + \frac{k_2}{2}\int_0^{\pi}(1-\cos(2\theta))\,d\theta$$

$$= \frac{k_1}{2}\left(\theta + \frac{\sin(2\theta)}{2}\right) + \frac{k_2}{2}\left(\theta - \frac{\sin(2\theta)}{2}\right)\Big|_0^{\pi}$$

$$= \frac{k_1}{2}\left(\pi \quad \frac{1}{2}(k_1\pi + k_2\pi)\right) = \pi H$$

$$\text{so } H = \frac{1}{\pi}\int_0^{\pi} k_n\,d\theta$$

b: Since θ is an angle that corresponds to some direction, it follows for instance that $\theta + \frac{\pi}{2}$ is perpendicular to the original direction determined by θ.

$$k_n\left(\theta + \frac{\pi}{2}\right) = k_1\cos^2\left(\theta + \frac{\pi}{2}\right) + k_2\sin^2\left(\theta + \frac{\pi}{2}\right)$$

$$= k_1(-\sin(\theta))^2 + k_2(\cos\theta)^2$$

$$= k_1\sin^2\theta + k_2\cos^2\theta$$

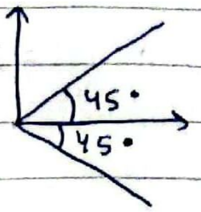
$$k_n(\theta) + k_n\left(\theta + \frac{\pi}{2}\right) = k_1\cos^2\theta + k_2\sin^2\theta + k_1\sin^2\theta + k_2\cos^2\theta$$

$= k_1 + k_2$ which does not depend on θ.

Q7 For a mean curvature to be zero, $H = \frac{k_1 + k_2}{2} = 0 \Rightarrow k_1 = -k_2$

$K = k_1 k_2 = -k_2^2 \leq 0$

$k_n(\theta) = 0 \iff k_1 \cos^2\theta = k_1 \sin^2\theta \iff \sin(\theta) = \pm\cos(\theta) \Rightarrow \theta = \frac{\pi}{4}$ or $\theta = \frac{-\pi}{4}$



Q8 (a) $z = x^2 + y^2$; $f(x,y,z) = x^2 + y^2 - z^2$; $S = \{(x,y,z) \in \mathbb{R}^3;\ f(x,y,z) = 0\}$

$$N(x,y,z) = \frac{\nabla f}{|\nabla f|} = \frac{(2x, 2y, -1)}{\sqrt{4(x^2+y^2)+1}}$$

(b) $x^2 + y^2 - z^2 = 1$; $S = \{(x,y,z) \in \mathbb{R}^3;\ f(x,y,z) = 0\}$ $f(x,y,z) = x^2 + y^2 - z^2 - 1$

$$N(x,y,z) = \frac{\nabla f}{|\nabla f|} = \frac{(x, y, -z)}{\sqrt{x^2+y^2+z^2}}$$

(c) $$N(x,y,z) = \frac{\nabla f}{|\nabla f|} = \frac{(2x, 2y, -2\cosh(z)\sinh(z))}{\sqrt{4(x^2+y^2+\cosh^2(z)\sinh^2(z))}}$$

Q9

(a) Let's assume for the sake of contradiction that the spherical image $N \circ \alpha$ is not a regular curve on S^2

For a curve to not be regular its tangent vectors must be zero at some point. So, there must exist a time t_0 for which derivative of spherical image equals zero: $(N \circ \alpha)'(t_0) = 0$

$(N \circ \alpha)'(t_0) = 0 \Rightarrow dN_{\alpha(t_0)}(\alpha'(t_0)) = 0$

That means there is a non-trivial α' $t_0 \in I$ such that $\alpha'(t_0) \neq 0$ which means that the null space of dN is ~~emp~~ non-empty and hence $\det(dN_p) = 0$ so $k_1 k_2 = 0$ which means it must contain planar ($k_1 = k_2 = 0$) or parabolic (k one zero, other non-zero)

Therefore we reach a contradiction as we had no planar or parabolic points in the original statements. So therefore at all points $\alpha(t)$ both principal curvatures are non-zero

b) Consider the parametrization of ~~arc~~ curve α by arc length s. Let $\beta = N \circ \alpha$ be the spherical image of C. Since C is line of curvature, it follows that $\beta'(s) = dN\left(\frac{d\alpha}{ds}\right) = \lambda(s)\alpha'(s)\ \forall s \in I$

$$\beta''(s) = \lambda'(s)\alpha'(s) + \lambda(s)\alpha''(s)$$

$$k_N(s) = \frac{|\beta'(s) \wedge \beta''(s)|}{|\beta'(s)|^3} = \frac{|\lambda(s)|^2\,|\alpha'(s) \wedge \alpha''(s)|}{|\lambda(s)|^3\,|\alpha'(s)|^3} \quad |\alpha'(s)| = 1$$

$$= \frac{1}{|\lambda(s)|}\,|\alpha'(s) \wedge \alpha''(s)| = \frac{k(s)}{|\lambda(s)|}$$

$$k_N(\alpha'(s)) = \mathrm{II}_{\alpha(s)}(\alpha'(s)) = -\langle dN_{\alpha(s)}(\alpha'(s)), \alpha'(s)\rangle = -\lambda(s)\langle\alpha'(s), \alpha'(s)\rangle$$
$$= -\lambda(s)$$

$$|k_n \alpha'(s)| = |\lambda(s)|$$
$$k = |\lambda| K_n = |k_n| K_n = |k_n K_n|$$

Q10: Let $C = \alpha(I)$ be a line of curvature on the surface S. $\alpha(s)$ is parametrized by arc length so $|\alpha'(s)| = 1$. Also $dN_{\alpha(s)}(\alpha'(s)) = \lambda(s)\alpha'(s)\ \forall s \in I$.
If α is a line of curvature it's tangent $t(s) = \alpha'(s)$ is an eigenvector of the ~~Gr~~ shape operator dN_p
Since C's nowhere tangent to an asymptotic direction, normal curvature k_n of S along $t(s)$ is never zero. The angle spanned by the plane of C (spanned by $t(s)$ and $n(s)$) and the tangent plane of S is constant.

The angle between two planes is determined by the angle between their normal vectors
Normal vector to C is the osculating plane & tangent plane to S at $\alpha(s)$ is $T_{\alpha(s)}S$.

$$\cos\theta = \frac{\langle b(s), N(\alpha(s))\rangle}{|b(s)|\,|N(\alpha(s))|} = \text{constant} \rightarrow \langle b(s), N(\alpha(s))\rangle = \text{constant}$$

differentiating b/c: $\langle b'(s), N(\alpha(s))\rangle + \langle b(s), dN_{\alpha(s)}(\alpha'(s))\rangle = 0$
$\Leftrightarrow \langle \tau(s) n(s), N(\alpha(s))\rangle + \langle b(s), \lambda(s)\alpha'(s)\rangle = 0$
$\tau(s)\langle n(s), N(\alpha(s))\rangle = 0 \Rightarrow \tau(s) = 0$
Why can't $\langle n(s), N(\alpha(s))\rangle = 0$? Since C lies on a tangent to an asymptotic line, $k_n \neq 0\ \forall s \in I$

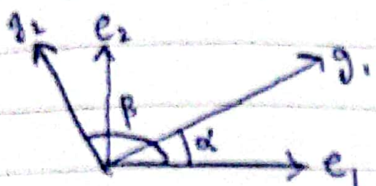
$k_n|\alpha'(s)| = k(s)\langle n(s), N(\alpha(s))\rangle \neq 0 \ \forall s \in I. \Rightarrow \langle n(s), N(\alpha(s))\rangle \neq 0$

Q) if r and r' are conjugates, then we know any conjugates of r and r' are also conjugates Consider: ① $g_1 = e_1\cos(\alpha) + e_2\sin(\alpha) \in r$

② $g_2 = e_1\cos(\beta) + e_2\sin(\beta) \in r'$

$\langle dN_p(g_1), g_2\rangle = 0 \iff -k_1\cos\alpha\cos\beta - k_2\sin\alpha\sin\beta = 0$

$\iff \tan\alpha\tan\beta = \frac{-k_1}{k_2}$ (for $\alpha, \beta \neq \frac{\pi}{2}, \frac{3\pi}{2}$)



Angle between g_1 & g_2 is $\beta-\alpha$.

$$\tan(\beta-\alpha) = \frac{\tan(\beta)-\tan(\alpha)}{1+\tan(\beta)\tan(\alpha)} = \frac{\frac{-k_1}{k_2}\cot(\alpha) - \tan(\alpha)}{1 - \frac{k_1}{k_2}}$$

$$= \frac{1}{1-\frac{k_1}{k_2}}\left[\frac{-k_1}{k_2}\cot(\alpha) - \tan(\alpha)\right]$$

differentiating right side for minimum:

$$\frac{1}{1-\frac{k_1}{k_2}}\left[\frac{-k_1}{k_2}(-\operatorname{cosec}^2(\alpha)) - \sec^2(\alpha)\right] = 0$$

$$\iff \frac{k_1}{k_2}\operatorname{cosec}^2(\alpha) - \sec^2(\alpha) = 0$$

$$\iff \frac{k_1}{k_2}\left[\frac{1}{\sin^2(\alpha)}\right] - \frac{1}{\cos^2\alpha} = 0$$

$$\tan^2(\alpha) = \frac{k_1}{k_2} \iff \tan(\alpha) = \pm\sqrt{\frac{k_1}{k_2}}$$

$$\tan\beta = \frac{-k_1}{k_2} \Big/ \pm\sqrt{\frac{k_1}{k_2}} = \mp\sqrt{\frac{k_1}{k_2}}$$

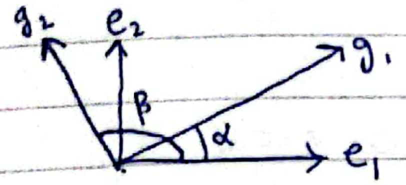
$k_n|\alpha'(s)| = k(s)\langle n(s), N(\alpha(s))\rangle \neq 0 \ \forall s \in I \Rightarrow \langle n(s), N(\alpha(s))\rangle \neq 0$

Q11 if r and r' are conjugates, then we know any conjugates of r and r' are also conjugates. Consider: ① $g_1 = e_1\cos(\alpha) + e_2\sin(\alpha) \in r$

② $g_2 = e_1\cos(\beta) + e_2\sin(\beta) \in r'$

$\langle dN_p(g_1), g_2\rangle = 0 \iff -k_1\cos\alpha\cos\beta - k_2\sin\alpha\sin\beta = 0$

$\iff \tan\alpha\tan\beta = \frac{-k_1}{k_2}$ (for $\alpha, \beta \neq \frac{\pi}{2}, \frac{3\pi}{2}$)



Angle between g_1 & g_2 is $\beta - \alpha$.

$$\tan(\beta-\alpha) = \frac{\tan(\beta) - \tan(\alpha)}{1 + \tan(\beta)\tan(\alpha)} = \frac{\frac{-k_1}{k_2}\cot(\alpha) - \tan(\alpha)}{1 - \frac{k_1}{k_2}}$$

$$= \frac{1}{1 - \frac{k_1}{k_2}}\left[\frac{-k_1}{k_2}\cot(\alpha) - \tan(\alpha)\right]$$

differentiating ~~right~~ right side for minimum:

$$\frac{1}{1 - \frac{k_1}{k_2}}\left[\frac{-k_1}{k_2}(-\operatorname{cosec}^2(\alpha)) - \sec^2(\alpha)\right] = 0$$

$$\iff \frac{k_1}{k_2}\operatorname{cosec}^2(\alpha) - \sec^2(\alpha) = 0$$

$$\iff \frac{k_1}{k_2}\left[\frac{1}{\sin^2(\alpha)}\right] - \cancel{\cos}\frac{1}{\cos^2\alpha} = 0$$

$$\tan^2(\alpha) = \frac{k_1}{k_2} \iff \tan(\alpha) = \pm\sqrt{\frac{k_1}{k_2}}$$

$$\tan\beta = \frac{-k_1}{k_2} \Big/ \pm\sqrt{\frac{k_1}{k_2}} = \mp\sqrt{\frac{k_1}{k_2}}$$

Let C be the asymptotic curve and $\alpha : I \to C$ be a parametrisation arc length. ~~We have~~ $\cancel{0 = k_n(\alpha'(s)) = k_1 \cos^2\theta_s + k_2 \sin^2\theta_s, \forall}$

13: $0 = k_n(\alpha'(s)) = k_1 \cos^2\theta_s + k_2 \sin^2\theta_2,\ \forall s \in I$, where k_1 & k_2 are the principal curvatures and θ is the angle between $\alpha'(s)$ & e_1

we have $k_1 \cos^2\theta_s = -k_2 \sin^2\theta_s,\ \forall s \in I$

$N'(s) = dN_{\alpha(s)}(\alpha'(s)) = dN_{\alpha(s)}(e_1 \cos\theta_s + e_2 \sin\theta_2)$

$= \cos\theta_s\, dN_{\alpha(s)}(e_1) + \sin\theta_s\, dN_{\alpha(s)}(e_2) = -k_1 \cos\theta_s e_1 - k_2 \sin\theta_s e_2$

$\|N'(s)\|^2 = k_1^2 \cos^2\theta_s + k_2 \sin^2\theta_s,\ \forall s \in I$

$\langle N(s), \alpha'(s)\rangle = 0 \iff \langle N'(s), \alpha'(s)\rangle + \langle N(s), \alpha''(s)\rangle = 0$

$\langle N(s), \alpha''(s)\rangle = 0, \Rightarrow -\langle dN_{\alpha(s)}(\alpha'(s)), \alpha'(s)\rangle$

$= II_{\alpha(s)}(\alpha'(s)) = k_n(\alpha'(s)) = 0$

$\langle N(s), \alpha''(s)\rangle = 0 \Rightarrow k(s)\langle N(s), n(s)\rangle = 0,\ \forall s \in I$

$N(s) \perp n(s)$ & $N(s) \perp \alpha'(s)$

So $N(s) = \pm b(s) = \pm J(s) n(s)$

$\|N'(s)\| = |J(s)|\, \|n(s)\| = |J(s)| \Rightarrow \|N'(s)\|^2 = |J(s)|^2$

$|J(s)|^2 = k_1 \cos^2\theta + k_2 \sin^2\theta \Rightarrow k_1(-k_2 \sin^2\theta) + k_2(-k_1 \cos^2\theta) = |J(s)|^2\ \forall,$

$-k_1 k_2(\sin^2\theta + \cos^2\theta) = |J(s)|^2$

$\cancel{|J(s)|^2 = -K \Rightarrow}\ |J(s)|^2 = -k_1 k_2 = -K \Rightarrow |J(s)| = \sqrt{-K}\ \forall s \in I$

14: